

CHAPTER K

ADDITIONAL REQUIREMENTS FOR HSS AND BOX-SECTION CONNECTIONS

This chapter addresses additional requirements for connections to HSS members and box sections of uniform wall thickness, where seam welds between box-section elements are complete-joint-penetration (CJP) groove welds in the connection region. The requirements of Chapter J also apply.

The chapter is organized as follows:

- K1. General Provisions and Parameters for HSS Connections
- K2. Concentrated Forces on HSS
- K3. HSS-to-HSS Truss Connections
- K4. HSS-to-HSS Moment Connections
- K5. Welds of Plates and Branches to HSS

K1. GENERAL PROVISIONS AND PARAMETERS FOR HSS CONNECTIONS

For the purposes of this chapter, the centerlines of branch members and chord members shall lie in a common plane. Rectangular HSS connections are further limited to having all members oriented with walls parallel to the plane.

The tables in this chapter are often accompanied by limits of applicability. Connections complying with the limits of applicability listed can be designed considering the limit states provided for each joint configuration. Connections not complying with the limits of applicability listed are not prohibited and must be designed by rational analysis.

User Note: The connection strengths calculated in Chapter K, including the applicable sections of Chapter J, are based on strength limit states only. See the Commentary if excessive connection deformations may cause serviceability or stability concerns.

User Note: Connection strength is often governed by the size of HSS members, especially the wall thickness of truss chords, and this must be considered in the initial design. To ensure economical and dependable connections can be designed, the connections should be considered in the design of the members. Angles between the chord and the branch(es) of less than 30° can make welding and inspection difficult and should be avoided. The limits of applicability provided reflect limitations on tests conducted to date, measures to eliminate undesirable limit states, and other considerations discussed in the Commentary. See Section J3.11b for through-bolt provisions.

This section provides parameters to be used in the design of plate-to-HSS and HSS-to-HSS connections.

The design strength, ϕR_n , ϕM_n , and ϕP_n , and the allowable strength, R_n/Ω , M_n/Ω , and P_n/Ω , of connections shall be determined in accordance with the provisions of this chapter and the provisions of Chapter B.

1. Definitions of Parameters

A_g = gross cross-sectional area of member, in.² (mm²)

B = overall width of rectangular HSS chord member measured 90° to the plane of the connection, in. (mm)

B_b = overall width of rectangular HSS branch member or plate measured 90° to the plane of the connection, in. (mm)

B_e = effective width of rectangular HSS branch member or plate for local yielding of the transverse element, in. (mm)

B_{ep} = effective width of rectangular HSS branch member or plate for punching shear, in. (mm)

D = outside diameter of round HSS chord member, in. (mm)

D_b = outside diameter of round HSS branch member, in. (mm)

F_c = available stress in chord member, ksi (MPa)

= F_y for LRFD; $0.60F_y$ for ASD

F_u = specified minimum tensile strength of HSS chord member material, ksi (MPa)

F_{ub} = specified minimum tensile strength of HSS branch member material, ksi (MPa)

F_y = specified minimum yield stress of HSS chord member material, ksi (MPa)

F_{yb} = specified minimum yield stress of HSS branch member or plate material, ksi (MPa)

H = overall height of rectangular HSS chord member measured in the plane of the connection, in. (mm)

H_b = overall height of rectangular HSS branch member measured in the plane of the connection, in. (mm)

Q_f = chord-stress interaction parameter

l_{end} = distance from the near side of the connecting branch or plate to end of chord, in. (mm)

t = design wall thickness of HSS chord member, in. (mm)

t_b = design wall thickness of HSS branch member or thickness of plate, in. (mm)

β = width ratio; the ratio of branch diameter to chord diameter = D_b/D for round HSS; the ratio of overall branch width to chord width = B_b/B for rectangular HSS

β_{eff} = effective width ratio; the sum of the perimeters of the two branch members in a K-connection divided by eight times the chord width

γ = chord slenderness ratio; the ratio of one-half the diameter to the wall thickness = $D/2t$ for round HSS, or the ratio of one-half the width to wall thickness = $B/2t$ for rectangular HSS

η = load length parameter, applicable only to rectangular HSS; the ratio of the length of contact of the branch with the chord in the plane of the connection to the chord width = l_b/B

θ = acute angle between the branch and chord, degrees

2. Rectangular HSS

2a. Effective Width for Connections to Rectangular HSS

For local yielding of transverse elements, the effective width of elements (plates or rectangular HSS branches) perpendicular to the longitudinal axis of a rectangular HSS member that deliver a force component transverse to the face of the member shall be taken as

$$B_e = \left(\frac{10}{B/t} \right) \left(\frac{F_y t}{F_y b t_b} \right) B_b \leq B_b \quad (\text{K1-1})$$

For shear yielding (punching), the effective width of the face of a rectangular HSS member, adjacent to the transverse element (plates or rectangular HSS branches) shall be taken as

$$B_{ep} = \left(\frac{10}{B/t} \right) B_b \leq B_b \quad (\text{K1-2})$$

User Note: Section J4 addresses the strength of affected elements in tension, compression, flexure, and shear. The effective widths defined in this section are used to establish the effective areas to be used when checking these limit states. The Commentary provides further guidance.

3. Chord-Stress Interaction Parameter

Where required, the chord-stress interaction parameter, Q_f , shall be taken as the following:

(a) For HSS chord member connecting surface in tension, $Q_f = 1$

(b) For round HSS chord member connecting surface in compression

$$Q_f = 1 - 0.3U(1 + U) \leq 1.0 \quad (\text{K1-3})$$

(c) For rectangular HSS chord member connecting surface in compression

(1) For T-, Y-, cross-, and transverse plate connections

$$0.4 \leq Q_f = 1.3 - 0.4 \left(\frac{U}{\beta} \right) \leq 1.0 \quad (\text{K1-4})$$

(2) For gapped K-connections

$$0.4 \leq Q_f = 1.3 - 0.4 \left(\frac{U}{\beta_{eff}} \right) \leq 1.0 \quad (\text{K1-5})$$

(3) For longitudinal plate connections

$$Q_f = 1 - 0.3U(1 + U) \leq 1.0 \quad (\text{K1-3})$$

where

$$U = \left| \frac{P_{ro}}{F_c A_g} + \frac{M_{ro}}{F_c S} \right| \leq 1.0 \quad (\text{K1-6})$$

where P_{ro} and M_{ro} are determined in the HSS chord member on the side of the joint that has lower compression stress for round HSS and higher compression stress for rectangular HSS. P_{ro} and M_{ro} refer to required strengths in the HSS chord: $P_{ro} = P_u$ for LRFD, and P_a for ASD; $M_{ro} = M_u$ for LRFD, and M_a for ASD.

Limits of applicability:

$D/t \leq 50$ for round HSS T-, Y-, and K-connections

$D/t \leq 40$ for round HSS cross-connections

B/t and $H/t \leq 35$ for rectangular HSS gapped K-connections and T-, Y-, and cross-connections

$F_y \leq 52$ ksi (360 MPa)

$F_y / F_u \leq 0.8$ (Note: ASTM A500/A500M Grade C is acceptable.)

4. End Distance

The available strength of the connection in Chapters J and K assume a chord member with a minimum end distance, l_{end} , on both sides of a connection.

(a) For rectangular sections

$$l_{end} \geq B\sqrt{1-\beta}, \text{ for } \beta \leq 0.85 \quad (\text{K1-7})$$

(b) For round sections

$$l_{end} \geq D \left(1.25 - \frac{\beta}{2} \right) \quad (\text{K1-8})$$

When the connection occurs at a distance less than l_{end} from an unreinforced end of the chord, the available strength of the connection shall be reduced by 50%.

K2. CONCENTRATED FORCES ON HSS

1. Definitions of Parameters

l_b = bearing length of the load measured parallel to the axis of the HSS member (or measured across the width of the HSS in the case of loaded cap plates), in. (mm)

2. Round HSS

The available strength of plate-to-round HSS connections, within the limits in Table K2.1A, shall be determined as shown in Table K2.1.

3. Rectangular HSS

The available strength of connections to rectangular HSS with concentrated loads shall be determined based on the applicable limit states from Chapter J.

K3. HSS-TO-HSS TRUSS CONNECTIONS

HSS-to-HSS truss connections consist of one or more branch members directly welded to a chord that passes as a continuous element through the connection. Such connections shall be classified as follows:

TABLE K2.1
Available Strengths of Plate-to-Round HSS Connections

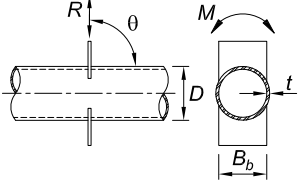
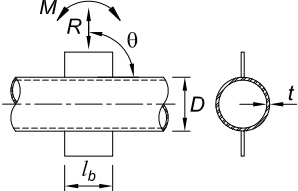
Connection Type	Connection Available Strength		Plate Bending
Transverse plate T-, Y-, and cross-connections 	Limit state: HSS local yielding		
	Plate axial load	In-plane	Out-of-plane
	$R_n \sin \theta = F_y t^2 \left[\frac{5.5}{1 - 0.81 \left(\frac{B_b}{D} \right)} \right] Q_f$ (K2-1a)	–	$M_n = 0.5 B_b R_n$ (K2-1b)
	$\phi = 0.90$ (LRFD) $\Omega = 1.67$ (ASD)		
Longitudinal plate T-, Y-, and cross-connections 	Limit state: HSS plastification		
	Plate axial load	In-plane	Out-of-plane
	$R_n \sin \theta = 5.5 F_y t^2 \left[1 + 0.25 \left(\frac{l_b}{D} \right) \right] Q_f$ (K2-2a)	$M_n = 0.8 l_b R_n$ (K2-2b)	–
	$\phi = 0.90$ (LRFD) $\Omega = 1.67$ (ASD)		

TABLE K2.1A
Limits of Applicability of Table K2.1

HSS wall slenderness:	$D/t \leq 50$ for T-connections under branch plate axial load or bending $D/t \leq 40$ for cross-connections under branch plate axial load or bending
Width ratio:	$0.2 < B_b/D \leq 1.0$ for transverse branch plate connections
Material strength:	$F_y \leq 52$ ksi (360 MPa)
Ductility:	$F_y/F_u \leq 0.8$ Note: ASTM A500/A500M Grade C is acceptable.

- (a) When the punching load, $P_r \sin \theta$, in a branch member is equilibrated by beam shear in the chord member, the connection shall be classified as a T-connection when the branch is perpendicular to the chord, and classified as a Y-connection otherwise.
- (b) When the punching load, $P_r \sin \theta$, in a branch member is essentially equilibrated (within 20%) by loads in other branch member(s) on the same side of the connection, the connection shall be classified as a K-connection. The relevant gap is between the primary branch members whose loads equilibrate.

User Note: A K-connection with one branch perpendicular to the chord is often called an N-connection.

- (c) When the punching load, $P_r \sin\theta$, is transmitted through the chord member and is equilibrated by branch member(s) on the opposite side, the connection shall be classified as a cross-connection.
- (d) When a connection has more than two primary branch members or branch members in more than one plane, the connection shall be classified as a general or multiplanar connection.

User Note: Limit states are not defined for general or multiplanar HSS-to-HSS truss connections.

When branch members transmit part of their load as K-connections and part of their load as T-, Y-, or cross-connections, the adequacy of the connections shall be determined by interpolation on the proportion of the available strength of each in total.

For trusses that are made with HSS that are connected by welding branch members to chord members, eccentricities within the limits of applicability are permitted without consideration of the resulting moments for the design of the connection.

1. Definitions of Parameters

$$O_v = l_{ov}/l_p \times 100, \%$$

e = eccentricity in a truss connection, positive being away from the branches, in. (mm)

g = gap between toes of branch members in a gapped K-connection, neglecting the welds, in. (mm)

$$l_b = H_b/\sin\theta, \text{ in. (mm)}$$

l_{ov} = overlap length measured along the connecting face of the chord beneath the two branches, in. (mm)

l_p = projected length of the overlapping branch on the chord, in. (mm)

ζ = gap ratio; the ratio of the gap between the branches of a gapped K-connection to the width of the chord = g/B for rectangular HSS

2. Round HSS

The available strength of round HSS-to-HSS truss connections, within the limits in Table K3.1A, shall be taken as the lowest value obtained according to the limit states shown in Table K3.1.

3. Rectangular HSS

The available strength, ϕP_n and P_n/Ω , of rectangular HSS-to-HSS truss connections within the limits in Table K3.2A, shall be taken as the lowest value obtained according to limit states shown in Table K3.2 and Chapter J.

User Note: Outside the limits in Table K3.2A, the limit states of Chapter J are still applicable and the applicable limit states of Chapter K are not defined.

User Note: Maximum gap size in Table K3.2A will be controlled by the e/H limit. If the gap is large, treat as two Y-connections.

TABLE K3.1

Available Strengths of Round
HSS-to-HSS Truss Connections

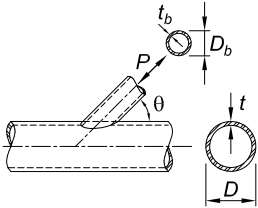
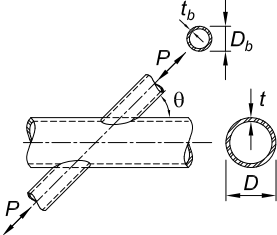
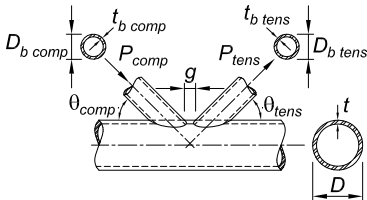
Connection Type	Connection Available Axial Strength
General check for T-, Y-, cross-, and K-connections with gap, when $D_b \text{ (tens/comp)} < (D - 2t)$	Limit state: shear yielding (punching) $P_n = 0.6 F_y t \pi D_b \left(\frac{1 + \sin \theta}{2 \sin^2 \theta} \right) \quad (\text{K3-1})$ $\phi = 0.95$ (LRFD) $\Omega = 1.58$ (ASD)
T- and Y-connections 	Limit state: chord plastification $P_n \sin \theta = F_y t^2 \left(3.1 + 15.6 \beta^2 \right) \gamma^{0.2} Q_f \quad (\text{K3-2})$ $\phi = 0.90$ (LRFD) $\Omega = 1.67$ (ASD)
Cross-connections 	Limit state: chord plastification $P_n \sin \theta = F_y t^2 \left(\frac{5.7}{1 - 0.81 \beta} \right) Q_f \quad (\text{K3-3})$ $\phi = 0.90$ (LRFD) $\Omega = 1.67$ (ASD)
K-connections with gap or overlap 	Limit state: chord plastification $(P_n \sin \theta)_{\text{compression branch}}$ $= F_y t^2 \left[2.0 + 11.33 \left(\frac{D_b \text{ comp}}{D} \right) \right] Q_g Q_f \quad (\text{K3-4})$ $(P_n \sin \theta)_{\text{tension branch}}$ $= (P_n \sin \theta)_{\text{compression branch}} \quad (\text{K3-5})$ $\phi = 0.90$ (LRFD) $\Omega = 1.67$ (ASD)
Functions	
$Q_g = \gamma^{0.2} \left[1 + \frac{0.024 \gamma^{1.2}}{\exp \left(\frac{0.5g}{t} - 1.33 \right) + 1} \right] \quad (\text{K3-6})$ Note that $\exp(x)$ is equal to e^x, where $e = 2.71828$ is the base of the natural logarithm.	

TABLE K3.1A
Limits of Applicability of Table K3.1

Connection eccentricity:	$-0.55 \leq e/D \leq 0.25$ for K-connections
Chord wall slenderness:	$D/t \leq 50$ for T-, Y-, and K-connections $D/t \leq 40$ for cross-connections
Branch wall slenderness:	$D_b/t_b \leq 50$ for tension and compression branch $D_b/t_b \leq 0.05E/F_{yb}$ for compression branch
Width ratio:	$0.2 \leq D_b/D \leq 1.0$ for T-, Y-, cross-, and overlapped K-connections $0.4 \leq D_b/D \leq 1.0$ for gapped K-connections
Gap:	$g \geq t_{b \text{ comp}} + t_{b \text{ tens}}$ for gapped K-connections
Overlap:	$25\% \leq O_v \leq 100\%$ for overlapped K-connections
Branch thickness:	$t_{b \text{ overlapping}} \leq t_{b \text{ overlapped}}$ for branches in overlapped K-connections
Material strength:	F_y and $F_{yb} \leq 52 \text{ ksi (360 MPa)}$
Ductility strength:	F_y/F_u and $F_{yb}/F_{ub} \leq 0.8$ Note: ASTM A500/A500M Grade C is acceptable.

User Note: The available axial strength for rectangular HSS-to-HSS member connections, ϕP_n or P_n/Ω , is obtained from Chapter J and the AISC *Steel Construction Manual* Part 9.

K4. HSS-TO-HSS MOMENT CONNECTIONS

HSS-to-HSS moment connections are defined as connections that consist of one or two branch members that are directly welded to a continuous chord that passes through the connection, with the branch or branches loaded by bending moments.

A connection shall be classified as

- (a) A T-connection when there is one branch and it is perpendicular to the chord and as a Y-connection when there is one branch, but not perpendicular to the chord
- (b) A cross-connection when there is a branch on each (opposite) side of the chord

1. Definitions of Parameters

Z_b = plastic section modulus of branch about the axis of bending, in.³ (mm³)

2. Round HSS

The available strength of round HSS-to-HSS moment connections within the limits of Table K4.1A shall be taken as the lowest value of the applicable limit states shown in Table K4.1.

TABLE K3.2
Available Strengths of Rectangular
HSS-to-HSS Truss Connections

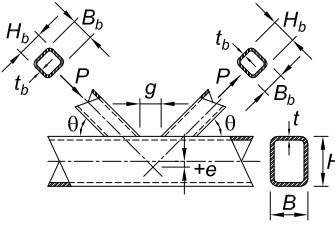
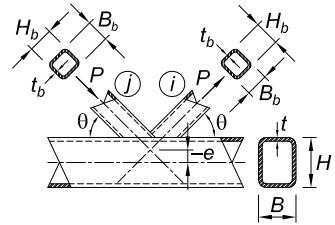
Connection Type	Connection Available Axial Strength
Gapped K-Connections 	Limit state: chord wall plastification, for all β $P_n \sin \theta = F_y t^2 (9.8 \beta_{eff} \gamma^{0.5}) Q_f \quad (K3-7)$ $\phi = 0.90 \text{ (LRFD)} \quad \Omega = 1.67 \text{ (ASD)}$
	Limit state: shear yielding (punching), when $B_b < B - 2t$ This limit state need not be checked for square branches. $P_n \sin \theta = 0.6 F_y t B (2\eta + \beta + \beta_{eop}) \quad (K3-8)$ $\phi = 0.95 \text{ (LRFD)} \quad \Omega = 1.58 \text{ (ASD)}$
	Limit state: shear of chord side walls in the gap region Determine $P_n \sin \theta$ in accordance with Section G4. This limit state need not be checked for square chords.
	Limit state: local yielding of branch/branches due to uneven load distribution This limit state need not be checked for square branches or where $B/t \geq 15$. $P_n = F_{yb} t_b (2H_b + B_b + B_e - 4t_b) \quad (K3-9)$ $\phi = 0.95 \text{ (LRFD)} \quad \Omega = 1.58 \text{ (ASD)}$
Overlapped K-Connections  Note that the force arrows shown for overlapped K-connections may be reversed; i and j control member identification.	Limit state: local yielding of branch/branches due to uneven load distribution $\phi = 0.95 \text{ (LRFD)} \quad \Omega = 1.58 \text{ (ASD)}$
	When $25\% \leq O_v < 50\%$ $P_{n,i} = F_{ybi} t_{bi} \left[\frac{O_v}{50} (2H_{bi} - 4t_{bi}) + B_{eoi} + B_{eov} \right] \quad (K3-10)$
	When $50\% \leq O_v < 80\%$ $P_{n,i} = F_{ybi} t_{bi} (2H_{bi} - 4t_{bi} + B_{eoi} + B_{eov}) \quad (K3-11)$
	When $80\% \leq O_v \leq 100\%$ $P_{n,i} = F_{ybi} t_{bi} (2H_{bi} - 4t_{bi} + B_{bi} + B_{eov}) \quad (K3-12)$ $B_{eoi} = \frac{10}{B/t} \left(\frac{F_y t}{F_{ybi} t_{bi}} \right) B_{bi} \leq B_{bi} \quad (K3-13)$ $B_{eov} = \frac{10}{B_{bj}/t_{bj}} \left(\frac{F_{ybj} t_{bj}}{F_{ybi} t_{bi}} \right) B_{bi} \leq B_{bi} \quad (K3-14)$ Subscript i refers to the overlapping branch. Subscript j refers to the overlapped branch. $P_{n,j} = P_{n,i} \left(\frac{F_{ybj} A_{bj}}{F_{ybi} A_{bi}} \right) \quad (K3-15)$
Functions	
$\beta_{eff} = \left[(B_b + H_b)_{compression\ branch} + (B_b + H_b)_{tension\ branch} \right] / 4B \quad (K3-16)$	
$\beta_{eop} = \frac{B_{ep}}{B} \quad (K3-17)$	

TABLE K3.2A
Limits of Applicability of Table K3.2

Connection eccentricity:	$-0.55 \leq e/H \leq 0.25$ for K-connections
Chord wall slenderness:	B/t and $H/t \leq 35$ for gapped K-connections and T-, Y-, and cross-connections $B/t \leq 30$ for overlapped K-connections $H/t \leq 35$ for overlapped K-connections
Branch wall slenderness:	B_b/t_b and $H_b/t_b \leq 35$ for tension branch $\leq 1.25 \sqrt{\frac{E}{F_{yb}}}$ for compression branch of gapped K-, T-, Y-, and cross-connections ≤ 35 for compression branch of gapped K-, T-, Y-, and cross-connections $\leq 1.1 \sqrt{\frac{E}{F_{yb}}}$ for compression branch of overlapped K-connections
Width ratio:	B_b/B and $H_b/B \geq 0.25$ for T-, Y-, cross-, and overlapped K-connections
Aspect ratio:	$0.5 \leq H_b/B_b \leq 2.0$ and $0.5 \leq H/B \leq 2.0$
Overlap:	$25\% \leq O_v \leq 100\%$ for overlapped K-connections
Branch width ratio:	$B_{bi}/B_{bj} \geq 0.75$ for overlapped K-connections, where subscript i refers to the overlapping branch and subscript j refers to the overlapped branch
Branch thickness ratio:	$t_{bi}/t_{bj} \leq 1.0$ for overlapped K-connections, where subscript i refers to the overlapping branch and subscript j refers to the overlapped branch
Material strength:	F_y and $F_{yb} \leq 52$ ksi (360 MPa)
Ductility:	F_y/F_u and $F_{yb}/F_{ub} \leq 0.8$ Note: ASTM A500/A500M Grade C is acceptable.
Additional Limits for Gapped K-Connections	
Width ratio:	B_b/B and $H_b/B \geq 0.1 + \frac{\gamma}{50}$ $\beta_{eff} \geq 0.35$
Gap ratio:	$\xi = g/B \geq 0.5(1 - \beta_{eff})$
Gap:	$g \geq t_{b \text{ compression branch}} + t_{b \text{ tension branch}}$
Branch size:	smaller $B_b \geq 0.63(\text{larger } B_b)$, if both branches are square

TABLE K4.1
Available Strengths of Round HSS-to-HSS
Moment Connections

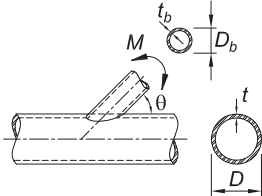
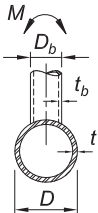
Connection Type	Connection Available Flexural Strength
Branch(es) under in-plane bending T-, Y-, and cross-connections 	Limit state: chord plastification $M_{n-ip} = 5.39F_y t^2 \gamma^{0.5\beta} \left(\frac{D_b}{\sin\theta} \right) Q_f \quad (K4-1)$ $\phi = 0.90$ (LRFD) $\Omega = 1.67$ (ASD)
	Limit state: shear yielding (punching), when $D_b < (D - 2t)$ $M_{n-ip} = 0.6F_y t D_b^2 \left(\frac{1 + 3\sin\theta}{4\sin^2\theta} \right) \quad (K4-2)$ $\phi = 0.95$ (LRFD) $\Omega = 1.58$ (ASD)
Branch(es) under out-of-plane bending T-, Y-, and cross-connections 	Limit state: chord plastification $M_{n-op} = \frac{F_y t^2 D_b}{\sin\theta} \left(\frac{3.0}{1 - 0.81\beta} \right) Q_f \quad (K4-3)$ $\phi = 0.90$ (LRFD) $\Omega = 1.67$ (ASD)
	Limit state: shear yielding (punching), when $D_b < (D - 2t)$ $M_{n-op} = 0.6F_y t D_b^2 \left(\frac{3 + \sin\theta}{4\sin^2\theta} \right) \quad (K4-4)$ $\phi = 0.95$ (LRFD) $\Omega = 1.58$ (ASD)
For T-, Y-, and cross-connections, with branch(es) under combined axial load, in-plane bending, and out-of-plane bending, or any combination of these load effects $\frac{P_r}{P_c} + \left(\frac{M_{r-ip}}{M_{c-ip}} \right)^2 + \frac{M_{r-op}}{M_{c-op}} \leq 1.0 \quad (K4-5)$ P_r = required axial strength in branch using LRFD or ASD load combinations, kips (N) M_{r-ip} = required in-plane flexural strength in branch using LRFD or ASD load combinations, kip-in. (N-mm) M_{r-op} = required out-of-plane flexural strength in branch using LRFD or ASD load combinations, kip-in. (N-mm) P_c = available axial strength obtained from Table K3.1, kips (N) M_{c-ip} = available strength for in-plane bending, kip-in. (N-mm) M_{c-op} = available strength for out-of-plane bending, kip-in. (N-mm)	

TABLE K4.1A
Limits of Applicability of Table K4.1

Chord wall slenderness:	$D/t \leq 50$ for T- and Y-connections $D/t \leq 40$ for cross-connections
Branch wall slenderness:	$D_b/t_b \leq 50$ $0.05E/F_{yb} \leq D_b/t_b$
Width ratio:	$0.2 < D_b/D \leq 1.0$
Material strength:	F_y and $F_{yb} \leq 52$ ksi (360 MPa)
Ductility:	F_y/F_u and $F_{yb}/F_{ub} \leq 0.8$ Note: ASTM A500/A500M Grade C is acceptable.

3. Rectangular HSS

The available strength, ϕP_n and P_n/Ω , of rectangular HSS-to-HSS moment connections within the limits in Table K4.2A shall be taken as the lowest value obtained according to limit states shown in Table K4.2 and Chapter J.

User Note: Outside the limits in Table K4.2A, the limit states of Chapter J are still applicable and the applicable limit states of Chapter K are not defined.

K5. WELDS OF PLATES AND BRANCHES TO HSS

The available strength of branch connections shall be determined considering the nonuniformity of load transfer along the line of weld, due to differences in relative stiffness of HSS walls in HSS-to-HSS connections and between elements in transverse plate-to-HSS connections, as follows:

$$R_n \text{ or } P_n = F_{nw} t_w l_e \tag{K5-1}$$

$$M_{n-ip} = F_{nw} S_{ip} \tag{K5-2}$$

$$M_{n-op} = F_{nw} S_{op} \tag{K5-3}$$

Interaction shall be considered.

(a) For fillet welds

$$\phi = 0.75 \text{ (LRFD)} \quad \Omega = 2.00 \text{ (ASD)}$$

(b) For partial-joint-penetration groove welds

$$\phi = 0.80 \text{ (LRFD)} \quad \Omega = 1.88 \text{ (ASD)}$$

TABLE K4.2
Available Strengths of Rectangular HSS-to-HSS
Moment Connections

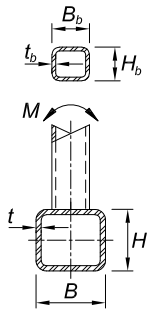
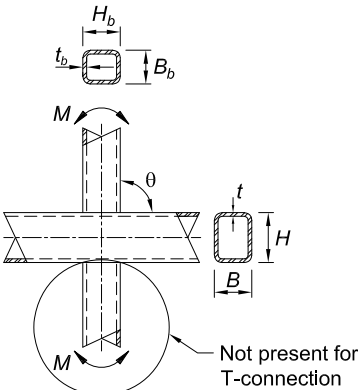
Connection Type	Connection Available Flexural Strength
Branch(es) under out-of-plane bending T- and cross-connections 	Limit state: chord sidewall local yielding $M_{n-op} = F_y^* t (B - t) (H_b + 5t) \quad (K4-6)$ $\phi = 1.00 \text{ (LRFD)} \quad \Omega = 1.50 \text{ (ASD)}$
	Limit state: chord distortional failure, for T-connections and unbalanced cross-connections $M_{n-op} = 2F_y t \left[H_b t + \sqrt{B H t (B + H)} \right] \quad (K4-7)$ $\phi = 0.95 \text{ (LRFD)} \quad \Omega = 1.58 \text{ (ASD)}$
Branch(es) under in-plane bending T- and cross-connections 	Limit state: sidewall local yielding When $\beta \geq 0.85$ $M_{n-ip} = 0.5F_y^* t (H_b + 5t)^2 \quad (K4-8)$ $\phi = 1.00 \text{ (LRFD)} \quad \Omega = 1.50 \text{ (ASD)}$
For T- and cross-connections, with branch(es) under combined axial load, in-plane bending, and out-of-plane bending, or any combination of these load effects $\frac{P_r}{P_c} + \frac{M_{r-ip}}{M_{c-ip}} + \frac{M_{r-op}}{M_{c-op}} \leq 1.0 \quad (K4-9)$ P_r = required axial strength in branch using LRFD or ASD load combinations, kips (N) M_{r-ip} = required in-plane flexural strength in branch using LRFD or ASD load combinations, kip-in. (N-mm) M_{r-op} = required out-of-plane flexural strength in branch using LRFD or ASD load combinations, kip-in. (N-mm) P_c = available axial strength, kips (N) M_{c-ip} = available strength for in-plane bending, kip-in. (N-mm) M_{c-op} = available strength for out-of-plane bending, kip-in. (N-mm) = ϕM_{n-op} (LRFD); = M_{n-op} / Ω (ASD)	
Functions	
$F_y^* = F_y$ for T-connections and $0.8F_y$ for cross-connections $P_{ro} = P_u$ for LRFD, and P_a for ASD; $M_{ro} = M_u$ for LRFD, and M_a for ASD	

TABLE K4.2A
Limits of Applicability of Table K4.2

Branch angle:	$\theta \cong 90^\circ$
Chord wall slenderness:	B/t and $H/t \leq 35$
Branch wall slenderness:	B_b/t_b and $H_b/t_b \leq 35$ $\leq 1.25 \sqrt{\frac{E}{F_{yb}}}$
Width ratio:	$B_b/B \geq 0.25$
Aspect ratio:	$0.5 \leq H_b/B_b \leq 2.0$ and $0.5 \leq H/B \leq 2.0$
Material strength:	F_y and $F_{yb} \leq 52 \text{ ksi (360 MPa)}$
Ductility:	F_y/F_u and $F_{yb}/F_{ub} \leq 0.8$ Note: ASTM A500/A500M Grade C is acceptable.

where

- F_{nw} = nominal stress of weld metal in accordance with Chapter J, ksi (MPa)
- S_{ip} = effective elastic section modulus of welds for in-plane bending (Table K5.1), in.³ (mm³)
- S_{op} = effective elastic section modulus of welds for out-of-plane bending (Table K5.1), in.³ (mm³)
- l_e = total effective weld length of groove and fillet welds to HSS for weld strength calculations, in. (mm)
- t_w = smallest effective weld throat thickness around the perimeter of branch or plate, in. (mm)

User Note: Where flexure results in tension in any load case in the weld the directional strength increase factor cannot exceed 1.0 in fillet welds to the end of rectangular HSS.

When a rectangular overlapped K-connection has been designed in accordance with Table K3.2, and the branch member component forces normal to the chord are 80% balanced (in other words, the branch member forces normal to the chord face differ by no more than 20%), the hidden weld under an overlapping branch may be omitted if the remaining welds to the overlapped branch everywhere develop the full capacity of the overlapped branch member walls.

The weld checks in Tables K5.1 and K5.2 are not required if the welds are capable of developing the full strength of the branch member wall along its entire perimeter (or a plate along its entire length).

User Note: The approach used here to allow downsizing of welds assumes a constant weld size around the full perimeter of the HSS branch. Special attention is required for equal width (or near-equal width) connections to rectangular HSS, which combine partial-joint-penetration groove welds along the matched edges of the connection, with fillet welds generally across the chord member face.

TABLE K5.1
Effective Weld Properties for Connections to Rectangular HSS

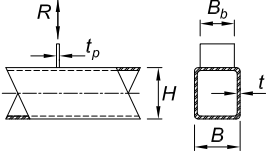
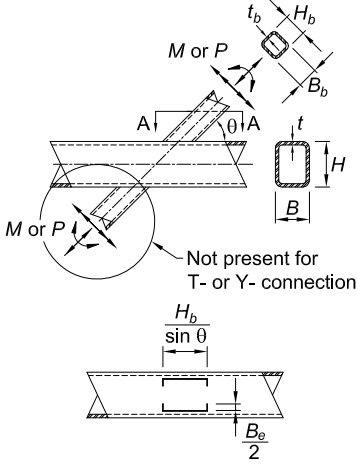
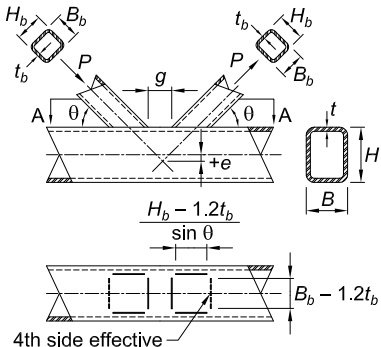
Connection Type	Weld Properties
Transverse plate T- and cross-connections under plate axial load 	Effective weld properties $l_e = 2B_e \quad (K5-4)$ where l_e = total effective weld length for welds on both sides of the transverse plate, in. (mm)
T-, Y-, and cross-connections under branch axial load or bending  Section A-A: Effective weld	Effective weld properties $l_e = \frac{2H_b}{\sin \theta} + 2B_e \quad (K5-5)$ $S_{ip} = \frac{t_w}{3} \left(\frac{H_b}{\sin \theta} \right)^2 + t_w B_e \left(\frac{H_b}{\sin \theta} \right) \quad (K5-6)$ $S_{op} = t_w \left(\frac{H_b}{\sin \theta} \right) B_b + \frac{t_w}{3} (B_b^2) - \frac{(t_w/3)(B_b - B_e)^3}{B_b} \quad (K5-7)$ When $\beta > 0.85$ or $\theta > 50^\circ$, $B_e/2$ shall not exceed $B_b/4$.
Gapped K-connections under branch axial load  Section A-A: Effective weld for $\theta \geq 60^\circ$	Effective weld properties When $\theta \leq 50^\circ$ $l_e = \frac{2(H_b - 1.2t_b)}{\sin \theta} + 2(B_b - 1.2t_b) \quad (K5-8)$ When $\theta \geq 60^\circ$ $l_e = \frac{2(H_b - 1.2t_b)}{\sin \theta} + B_b - 1.2t_b \quad (K5-9)$ When $50^\circ < \theta < 60^\circ$, linear interpolation shall be used to determine l_e.

TABLE K5.1 (continued)
Effective Weld Properties for Connections to Rectangular HSS

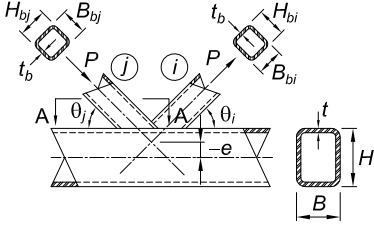
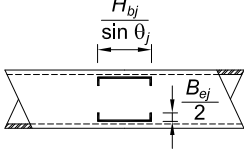
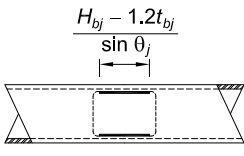
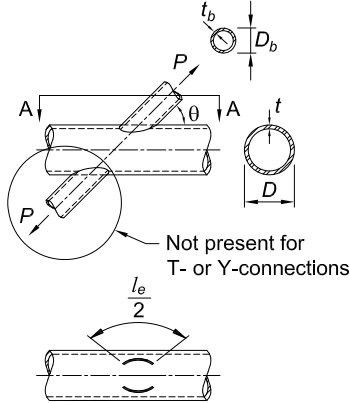
Connection Type	Weld Properties
Overlapped K-connections under branch axial load  Note that the force arrows shown for overlapped K-connections may be reversed; <i>i</i> and <i>j</i> control member identification.  Section A-A: Effective weld when $\frac{B_{bj}}{B} \leq 0.85$ and $\theta_j \leq 50^\circ$  Section A-A: Effective weld when $\frac{B_{bj}}{B} > 0.85$ or $\theta_j > 50^\circ$	Overlapping member effective weld properties (all dimensions are for the overlapping branch, <i>i</i>) When $25\% \leq O_v < 50\%$ $l_{e,i} = \frac{2O_v}{50} \left[\left(1 - \frac{O_v}{100} \right) \left(\frac{H_{bi}}{\sin \theta_i} \right) + \frac{O_v}{100} \left(\frac{H_{bi}}{\sin(\theta_i + \theta_j)} \right) \right] + B_{eoi} + B_{eov} \quad (K5-10)$ When $50\% \leq O_v < 80\%$ $l_{e,i} = 2 \left\{ \left(1 - \frac{O_v}{100} \right) \left(\frac{H_{bi}}{\sin \theta_i} \right) + \frac{O_v}{100} \left[\frac{H_{bi}}{\sin(\theta_i + \theta_j)} \right] \right\} + B_{eoi} + B_{eov} \quad (K5-11)$ When $80\% \leq O_v < 100\%$ $l_{e,i} = 2 \left\{ \left(1 - \frac{O_v}{100} \right) \left(\frac{H_{bi}}{\sin \theta_i} \right) + \frac{O_v}{100} \left(\frac{H_{bi}}{\sin(\theta_i + \theta_j)} \right) \right\} + B_{bi} + B_{eov} \quad (K5-12)$ $B_{eoi} = \frac{10}{B/t} \left(\frac{F_y t}{F_{ybi} t_{bi}} \right) B_{bi} \leq B_{bi} \quad (K3-13)$ $B_{eov} = \frac{10}{B_{bj}/t_{bj}} \left(\frac{F_{ybj} t_{bj}}{F_{ybi} t_{bi}} \right) B_{bi} \leq B_{bi} \quad (K3-14)$ When $B_{bi}/B > 0.85$ or $\theta_i > 50^\circ$, $B_{eoi}/2$ shall not exceed $B_{bi}/4$ and when $B_{bi}/B_{bj} > 0.85$ or $(180 - \theta_i - \theta_j) > 50^\circ$, $B_{eov}/2$ shall not exceed $B_{bi}/4$. Subscript <i>i</i> refers to the overlapping branch. Subscript <i>j</i> refers to the overlapped branch. Overlapped member effective weld properties (all dimensions are for the overlapped branch, <i>j</i>) $l_{e,j} = \frac{2H_{bj}}{\sin \theta_j} + 2B_{ej} \quad (K5-13)$ $B_{ej} = \frac{10}{B/t} \left(\frac{F_y t}{F_{ybj} t_{bj}} \right) B_{bj} \leq B_{bj} \quad (K5-14)$ When $B_{bj}/B > 0.85$ or $\theta_j > 50^\circ$ $l_{e,j} = 2(H_{bj} - 1.2t_{bj})/\sin \theta_j \quad (K5-15)$

TABLE K5.2
Effective Weld Properties for Connections to Round HSS

Connection Type	Weld Properties
T-, Y-, and cross-connections under branch axial load  Not present for T- or Y-connections Section A-A: Effective weld	Effective weld properties When $0.1 \leq \beta \leq 0.5$, $60^\circ \leq \theta \leq 90^\circ$, and $10 \leq D/t \leq 50$ $l_e = \frac{4}{\sqrt{2\beta(D/t)}} l_w \leq l_w \tag{K5-16}$ where l_w is the total weld length around the branch. This may be obtained from 3D models of intersection cylinders, or from $l_w = \pi D_b \frac{1 + 1/\sin\theta}{2} \tag{K5-17}$